Module: Engineering Team Project

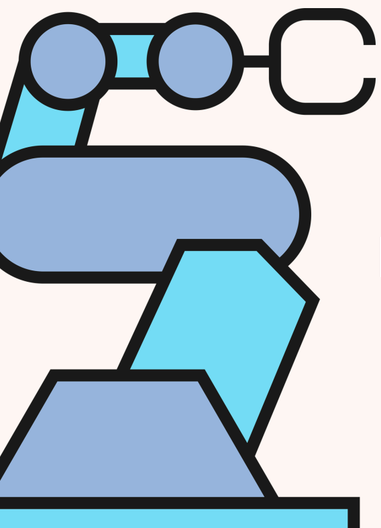
Development of a Arduino-Controlled Bipedal Robot

MEGATRON

MARY • ELON • GOURI • THIERRY

Project Aim & Team Roles

- Design and build a arduino controlled robot
- Enable wireless WiFi control
- Support multiple movement modes
- Carry additional weight
- Apply engineering project management techniques



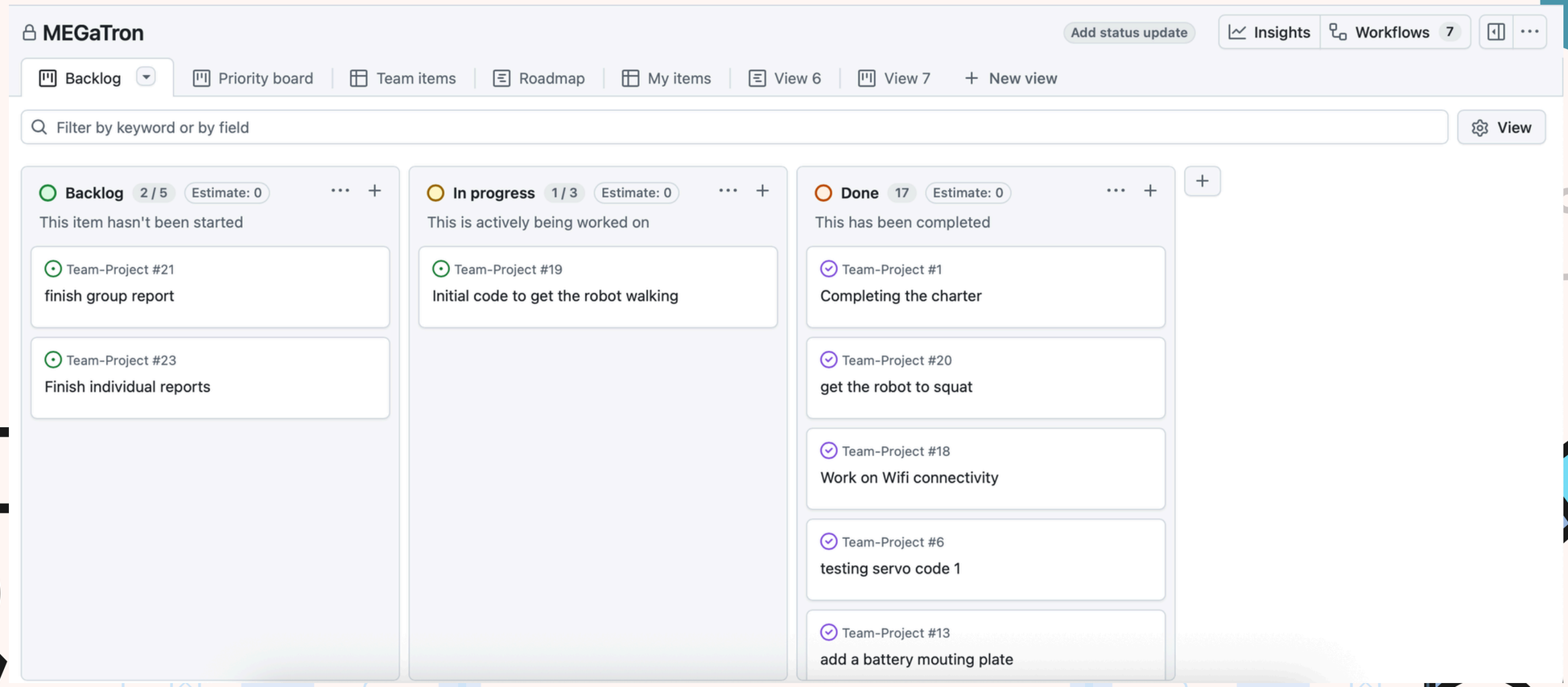
Gouri Girish Nair: Team Lead,Testing & Software

Elon Premroy: Mechanical Design & Hardware Lead

Thierry Baptiste: Hardware & Software Lead

Mary Ogunmoroti: Testing Lead, Documentation,Design & Software

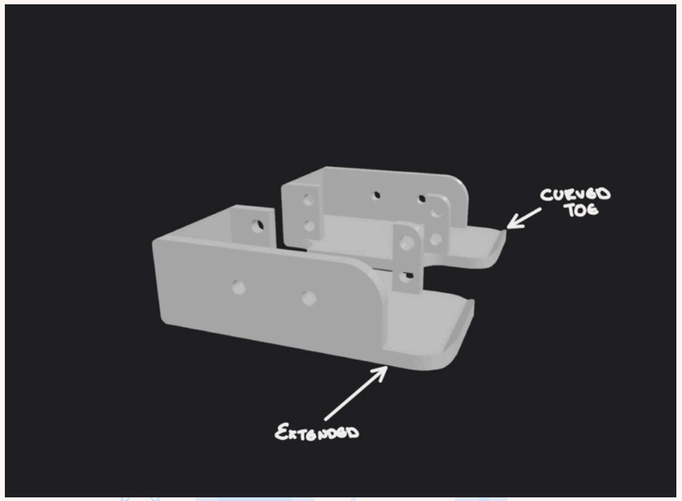
AGILE METHODOLOGY

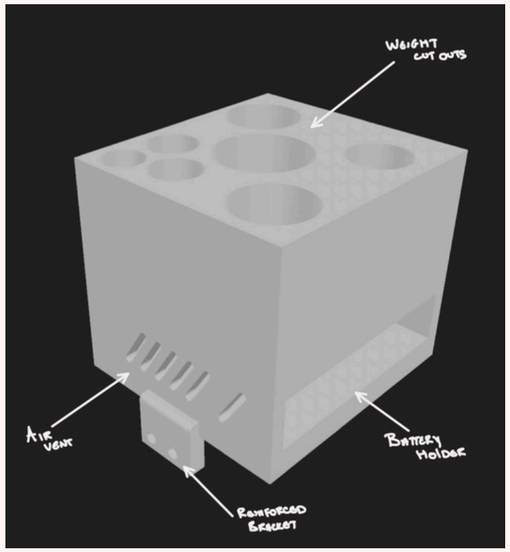


MECHANICAL DESIGN & HARDWARE

Redesigned Feet

- Created from the original foot geometry and modified to improve stability.
- Added curved front and rear edges to assist smoother weight transfer during movement.
- Increased overall strength around the servo mounting area.
- Initial prototype failed due to a thin servo connection point; the design was reinforced by increasing the material thickness around the mounting section.





Battery & Weight Housing

- Designed using the existing robot CAD model to retain correct servo and mounting dimensions.
- Included dedicated weight compartments to allow centre of gravity adjustments during testing.
- Integrated battery holder, ventilation slots, and reinforced mounting brackets.
- Battery holder brackets were widened after testing to improve strength and durability.

Software & Testing

- ESP32 used as the main controller
- WiFi Access Point created by ESP32
- Web-based control interface developed
- Commands were sent from a smartphone through a web interface hosted directly by the ESP32.
- Servo control routines implemented for standing, squatting and walking

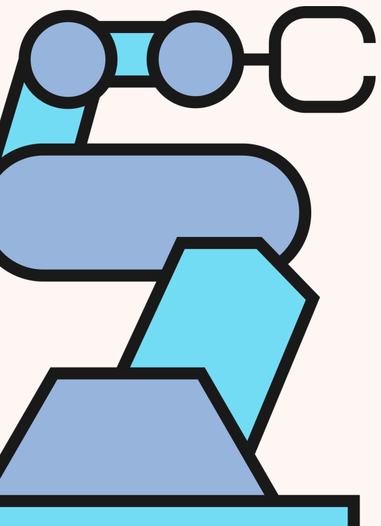
- Verified WiFi connectivity
- Tested webpage responsiveness
- Checked individual servo movements
- Evaluated standing and squat functions
- Conducted repeated trials to assess reliability

Challenges & Development Process



Key Challenges

- Maintaining balance during movement
- Achieving a stable squat motion
- Servo calibration and joint alignment
- Integrating WiFi control



How We Overcame These Challenges

- Repeated testing and calibration
- Mechanical adjustments and redesigns
- Refinement of servo control code
- Team collaboration and problem-solving

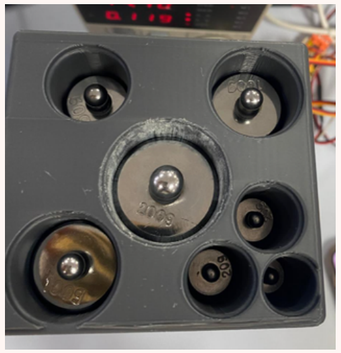
Results & Evaluation



Standing

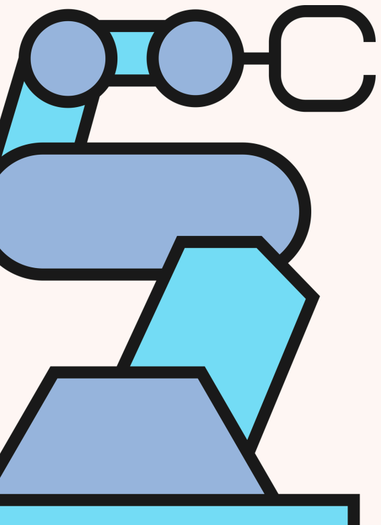


Squatting



Load Test

The robot successfully achieved standing, squatting, WiFi control and load support. Walking performance improved through repeated testing and calibration.



Future Improvements

- Improve walking stability
- Lower the centre of mass
- Add sensors for better balance control

One of the biggest lessons we learned was that small adjustments can make a big difference to robot performance. With more time, we would focus on improving balance and creating a smoother, more natural walking motion.

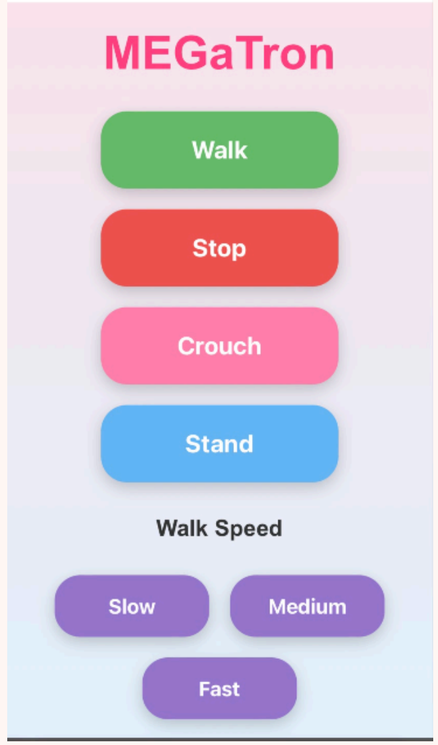
Final Robot Demonstration

Key Features Demonstrated

- Standing posture
- Squat functionality
- WiFi control via smartphone
- Load-bearing capability







By combining mechanical design, servo control, and wireless communication, the finished robot successfully demonstrated the main project goals.

Conclusion and Future Improvements

Conclusion

- Successfully designed, manufactured, and assembled a functioning bipedal robot.
- Integrated mechanical, electronic, and software systems into a single working platform.
- Developed wireless WiFi control for standing, crouching, and gait demonstrations.
- Improved robot stability through centre of gravity optimisation and redesigned feet.
- Gained valuable experience in CAD design, 3D printing, programming, testing, and systems integration.

Future Improvements

- Upgrade to higher torque servos to improve load carrying capability and walking performance.
- Further optimise the robot's centre of gravity and weight distribution.
- Develop more advanced walking algorithms for stable autonomous locomotion.
- Improve foot design to increase traction and reduce instability during movement.
- Add sensors such as an IMU or gyroscope to enable balance correction and autonomous control.
- Redesign the leg mechanism to provide greater freedom of movement and more natural walking behaviour.

Q&A

